

HYPERION: Constellation Optimization

Methodology

- Using STK Analyzer plugin and specifically, the Design of Experiments (DOE) Tool
- 3 Phase Strategy
 - Phase 1: Custom parameter sweep for LEO constellation
 - Phase 2: Custom parameter sweep for MEO constellation
 - Phase 3: Refinement and Verification for HYPERION constellation
- Phases 1 and 2 done simultaneously
- Phase 3 assimilates and verifies results
- 3 Phase strategy reduces search space from ~357,000,000 possibilities to ~27,000

Models and Assumptions

LEO & MEO Sats

- Circular orbit
- Single shell (multiple planes)
- J2 propagator
- Single day propagation (11/10/24 18:00:00 UTC - 11/11/24 18:00:00 UTC)
- Walker delta configuration (i: t/p/f)
- Cross-plane and in-plane visibility: Line of sight (LOS)

LEO Sensors

- Single SWIR
- Simple conic
- High resolution, large half angle
- Pointing along Nadir vector (to center of Earth)

MEO Sensors

- Single SWIR
- Simple conic
- Low resolution, small half angle
- Pointing along Nadir vector (to center of Earth)

Ground Coverage Definition

- +/- 90 deg latitude (full coverage)

- 10 deg point granularity

Ground Stations

- Near Space Network
- Visibility requirement: 5 deg minimum elevation angle
- LEOs route information in-plane to GS (max 3 hops)
- MEOS route information in-plane to GS (max 1 hops)
- Alaska Satellite Facility (ASFS): 64° 51' 31" N, 147° 51' 27" W
- NASA White Sands Facility (WSC): 32° 32' 27" N, 106° 36' 44" W
- NASA Wallops Ground Station (WGS): 37° 55' 28" N, 75° 28' 35" W

Missiles

- 3 missiles launched at intervals over the analysis window
- Modeled as aircraft with waypoints
- Ground launches only
- Launched from local terrain altitude
- Missile 1:
 - Launch site: Plesetsk Cosmodrome, Russia (62.96 N, 40.683 E)
 - Target site: White House, Washington D.C., U.S. (38.8977 N, 77.0365 W)
 - Time of Launch: 11/10/24 18:00:00 UTC
 - Time of Impact: 11/10/24 19:12:12.700 UTC
 - Speed: Mach 5
 - Altitude: 20 km
- Missile 2:
 - Launch site: Pacific Ocean (0 N, 160 E)
 - Target site: Military Ocean Terminal Concord, California, U.S. (37.9942 N, 121.983 W)
 - Time of Launch: 11/11/24 06:00:00 UTC
 - Time of Impact: 11/11/24 06:58:06.980 UTC
 - Speed: Mach 7.5
 - Altitude: 50 km
- Missile 3:
 - Launch site: Chaotanbi, China (20.0406 N, 110.765 E)
 - Target site: Naval Weapons Station Seal Beach, California, U.S. (33.75 N, 118.067 W)
 - Time of Launch: 11/11/24 15:00:00 UTC
 - Time of Impact: 11/11/24 15:58:54.010 UTC
 - Speed: Mach 10
 - Altitude: 80 km

Phase 1: LEO DOE

- Design variables (start, stop, step):
 - Inclination (80 deg, 90 deg, 2 deg)
 - Number of planes (4, 10, 1)
 - Sats per plane (3, 8, 1)
 - Phasing (2, 9, 1)
 - Altitude (500, 1000, 100)
 - SWIR conic half angle (30, 60, 10)
- Constraints:
 - $20 \leq \text{Number of planes} * \text{number of sats per plane} \leq 32$
 - $1 < \text{Phasing} < \text{Number of planes}$
- Outputs:
 - Overall maximum revisit time
 - Overall average revisit time
 - Missile 1 response time (time from launch to visibility)
 - Missile 2 response time
 - Missile 3 response time
 - Overall average missile access duration
- Weights for Scoring:
 - Overall maximum revisit time: 0.3
 - Overall average revisit time: 0.3
 - Average missile response time: 0.2
 - Average missile access duration: 0.2
- Search space: 8928 configurations

Phase 2: MEO DOE

- Design variables (start, stop, step):
 - Inclination (80 deg, 90 deg, 2 deg)
 - Number of planes (4, 10, 1)
 - Sats per plane (3, 8, 1)
 - Phasing (2, 9, 1)
 - Altitude (2000, 10000, 1000)
 - SWIR conic half angle (5, 25, 5)
- Constraints:
 - $20 \leq \text{Number of planes} * \text{number of sats per plane} \leq 32$
 - $1 < \text{Phasing} < \text{Number of planes}$
- Outputs:
 - Overall average access duration
 - Overall average revisit time

- Overall maximum revisit time
 - Missile 1 tracking time / Missile 1 flight time (percentage of visibility)
 - Missile 2 tracking time / Missile 2 flight time
 - Missile 3 tracking time / Missile 3 flight time
 - Overall average missile access duration
- Weights for Scoring:
 - Overall average access duration: 0.3
 - Overall average revisit time: 0.2
 - Overall maximum revisit time: 0.2
 - Total missile tracking time / total flight time: 0.2
 - Overall average missile access duration: 0.1
- Search space: 16740 configurations

Phase 3: HYPERION DOE

- Design table generated by synthesizing LEO and MEO DOE's
- Outputs:
 - Overall average revisit time
 - Overall maximum revisit time
 - Ground station to LEO sats to missiles total access duration
 - Ground station to MEO sats to missiles total access duration
 - Overall average LEO sat to ground station access
 - Overall average LEO sat to LEO sat access
 - Overall average LEO sat to MEO sat access
 - Overall average MEO sat to ground station access
 - Overall average MEO sat to MEO sat access
- Weights for Scoring:
 - Ground station to LEO sats to missiles total access: 0.1
 - Ground station to MEO sats to missiles total access: 0.1
 - Overall average LEO sat to ground station access: 0.15
 - Overall average LEO sat to LEO sat access: 0.1
 - Overall average LEO sat to MEO sat access: 0.3
 - Overall average MEO sat to ground station access: 0.15
 - Overall average MEO sat to MEO sat access: 0.1
- Search space: 625 configurations (top 25 LEO by top 25 MEO)
- Post-processing Verification:
 - No gaps in continuous LOS between LEO and MEO sats
 - No gaps in continuous LOS between LEO and LEO sats
 - No gaps in continuous LOS between MEO and MEO sats
 - Valid, continuous chains from LEO sats to ground stations
 - Valid, continuous chains from LEO sats to ground stations

- Average hops from LEO sats to ground stations [0, 3]
- Average hops from MEO sats to ground stations [0, 1]

Expected/Preliminary Results

- LEO completed trial count: 2142
- MEO completed trial count: 1362
- HYPERION completed trial count: 1 (top LEO by top MEO)

Design Variables

Variable	LEO Shell 1	LEO Shell 2	MEO
Inclination (i)	82.0°	10.0°	90.0°
Total number of sats (t)	30	2	21
Planes (p)	5	1	7
Phasing factor (f)	3	0	3
Altitude (h)	1000 km	1000 km	10000 km
Sensor conic half angle (η)	60°	60°	25°

*Second LEO shell added to aid with poor equatorial coverage

Performance

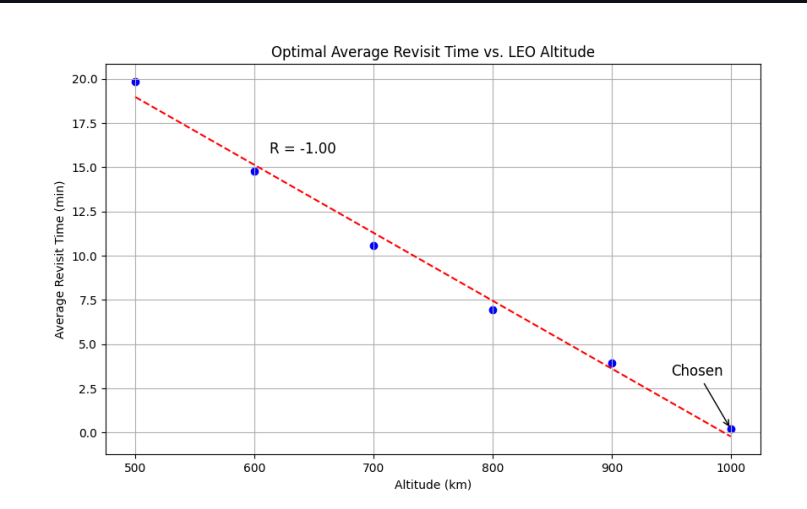
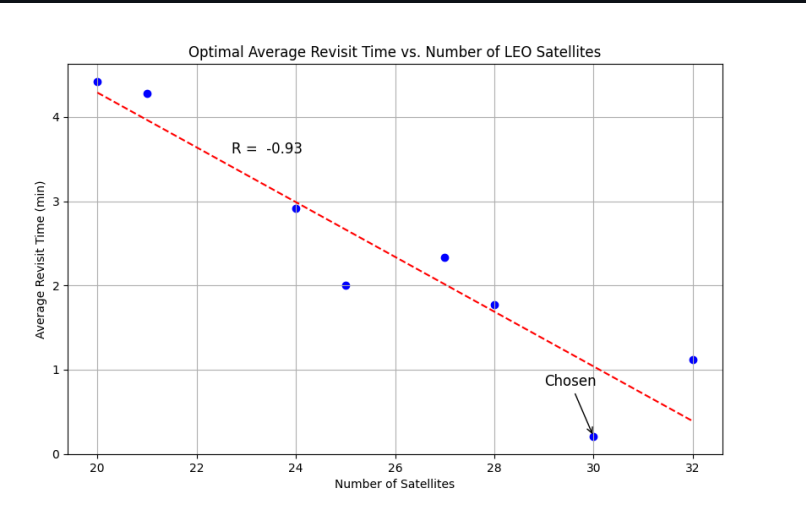
	LEO Performance	
	One Shell	Two Shells
Average revisit (min)	3.2087e-1	3.1517e-1
Maximum revisit (min)	1.6496	1.6463
Average missile response time (s)	1.1427	4.6667e-3
Average missile access duration (min)	7.7945	6.898

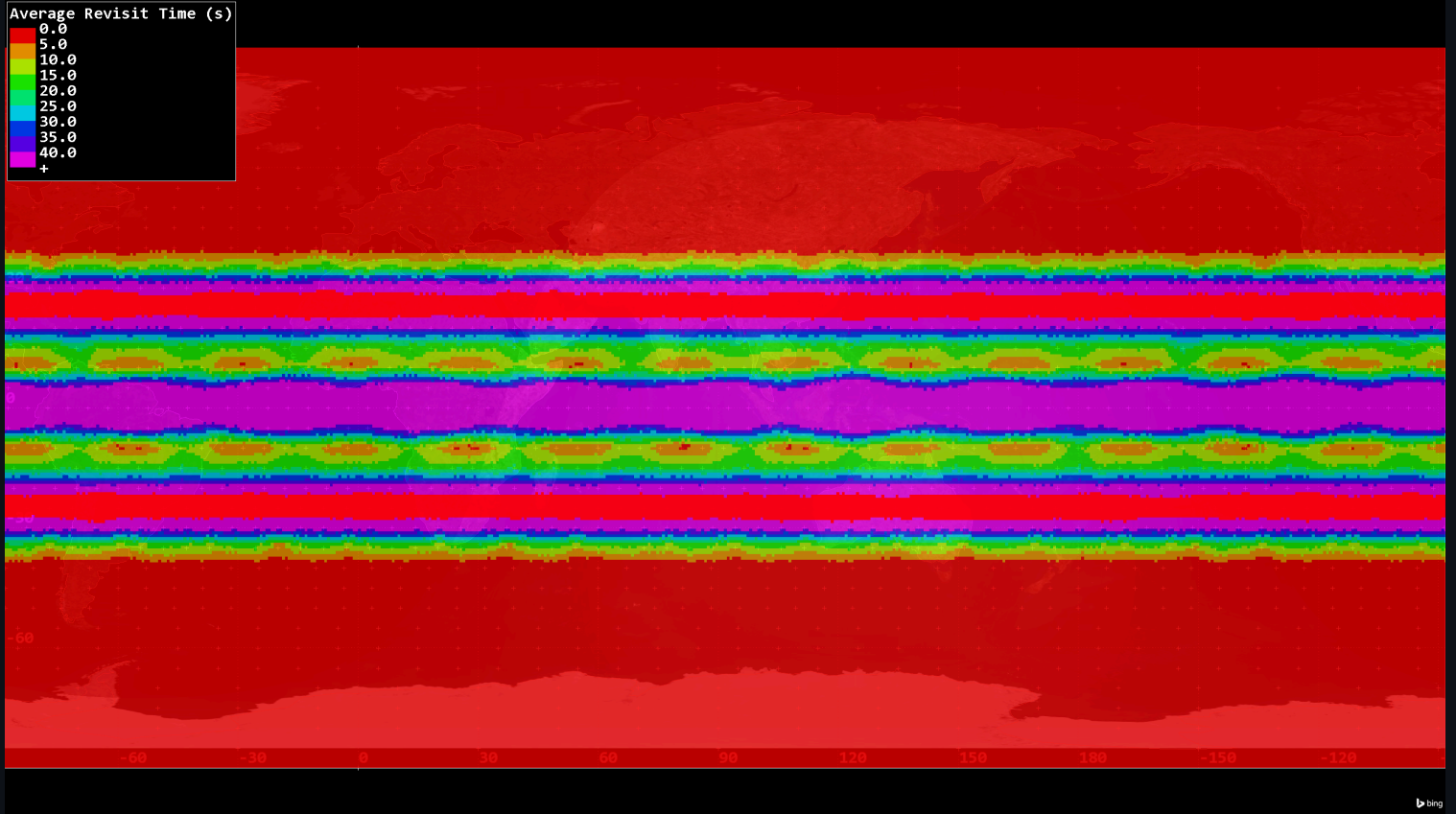
MEO Performance	
Average revisit (min)	0.0
Maximum revisit (min)	0.0
Average access duration (min)	105.30
% Missile visibility	100
Average missile access duration (min)	47.499

HYPERION Performance	
Gaps in LOS between LEO and MEO (boolean)	False
Gaps in LOS between LEO and LEO (boolean)	False
Gaps in LOS between MEO and MEO (boolean)	False
Average total gap time in valid chain between LEO and GS (min)	0.0
Average total gap time in valid chain between MEO and GS (min)	0.0
Average number of hops from LEO to ground stations	2.7610
Average number of hops from MEO to ground stations	0.9424
Average number of hops from missiles to LEO to ground stations	1.8274
Average number of hops from missiles to MEO to ground stations	0.9041

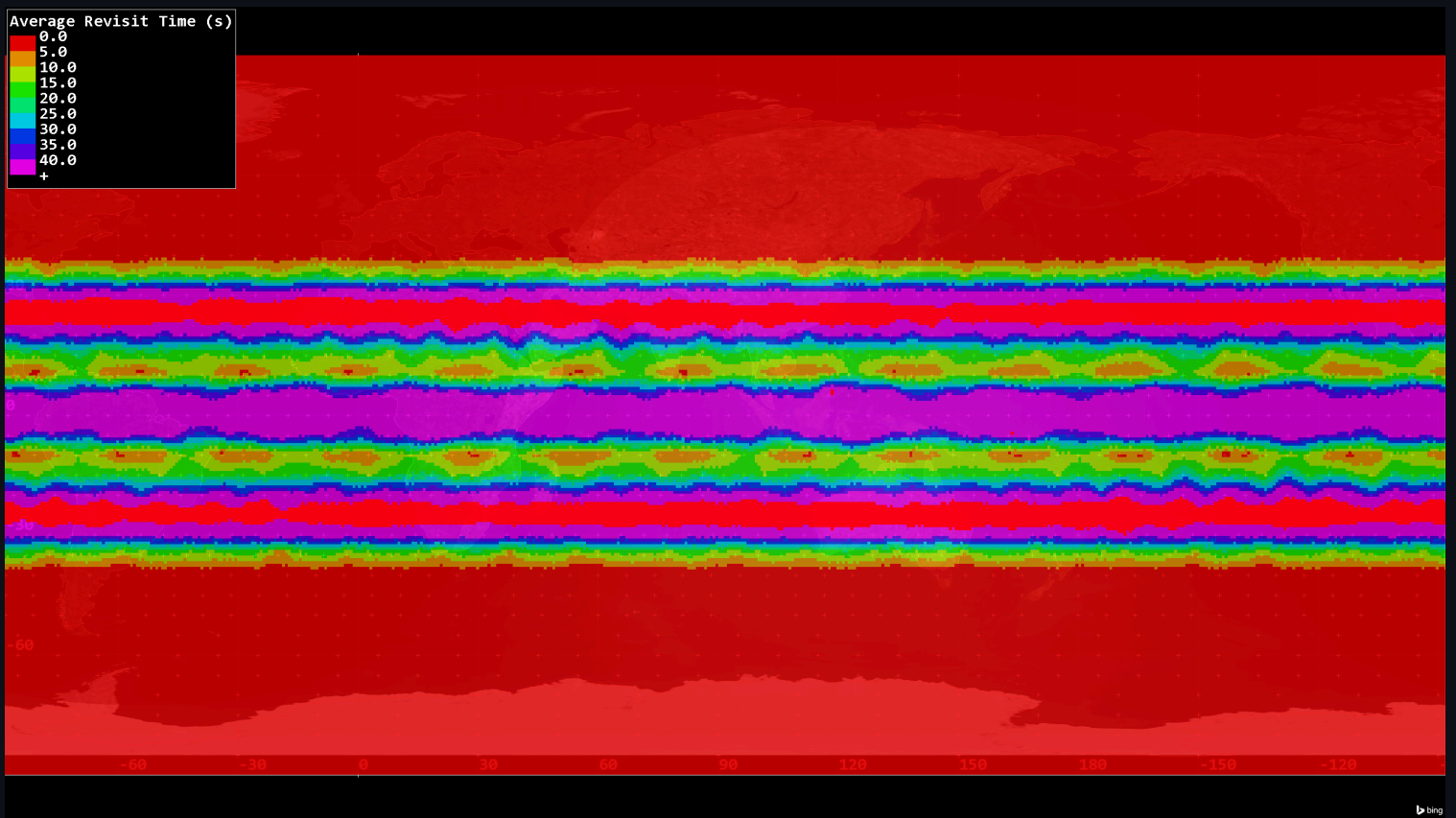
Figures

LEO



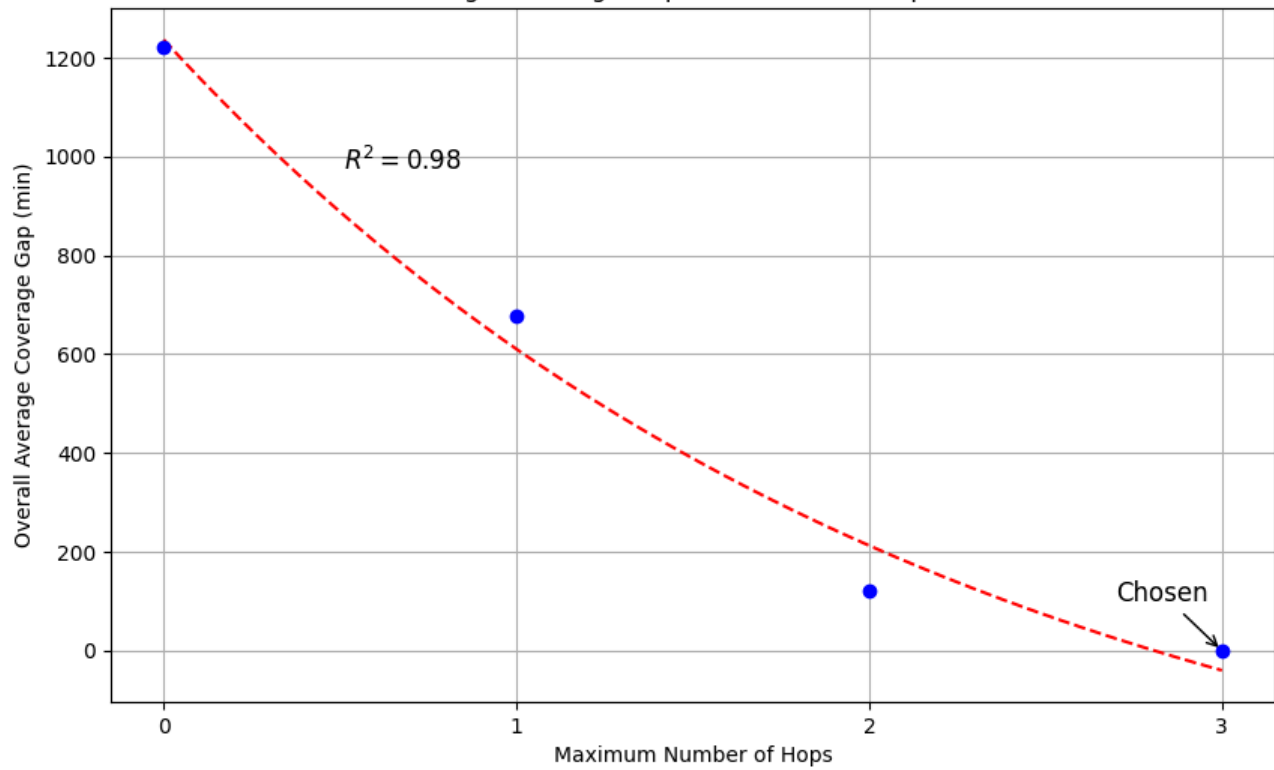


Average LEO revisit time heat map (single shell)



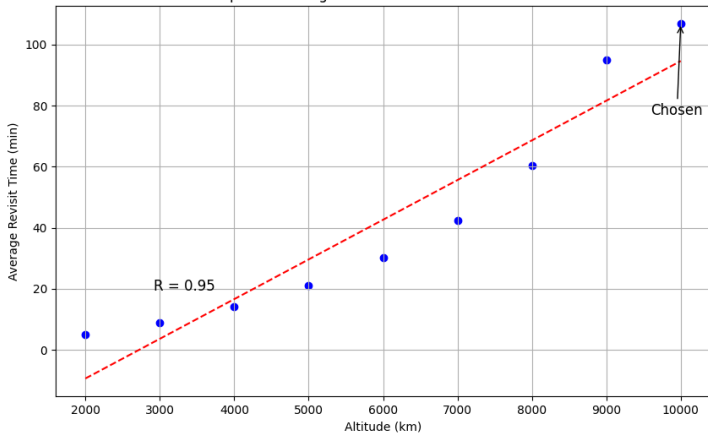
Average LEO revisit time heat map (two shells)

Overall Average Coverage Gaps vs. Maximum Hops (LEO to GS)

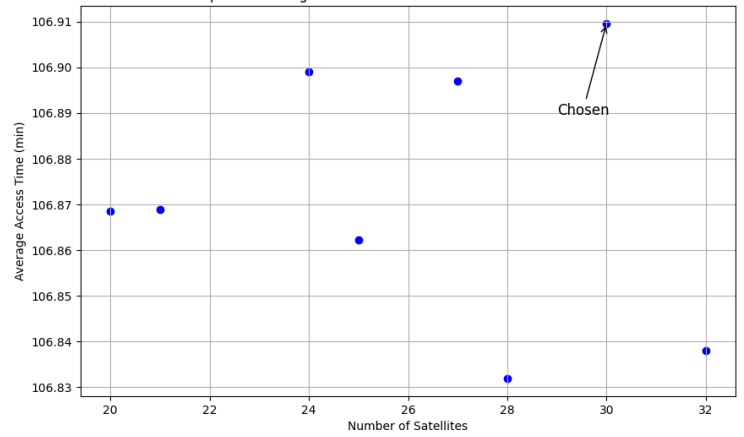


MEO

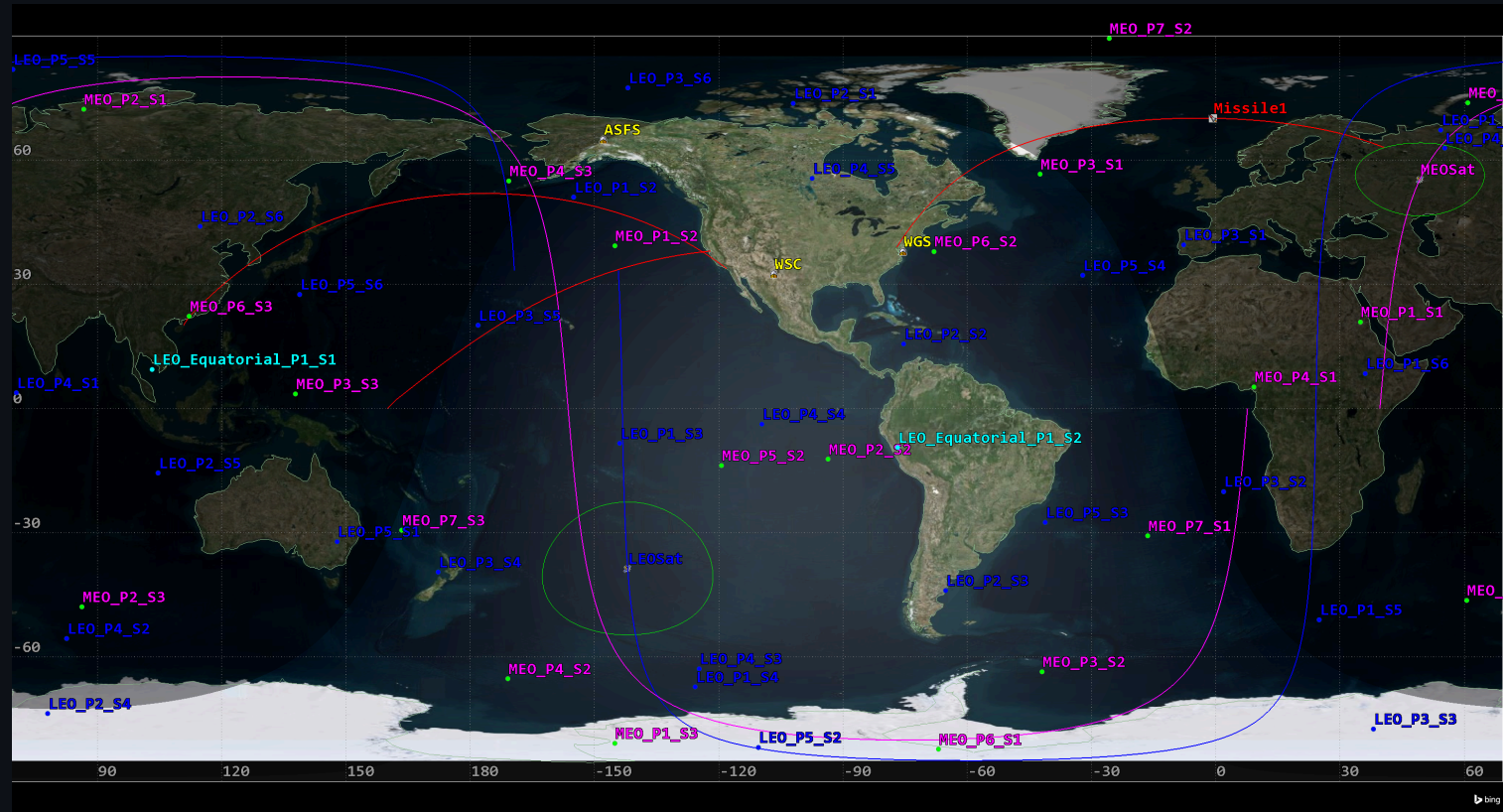
Optimal Average Access Time vs. MEO Altitude



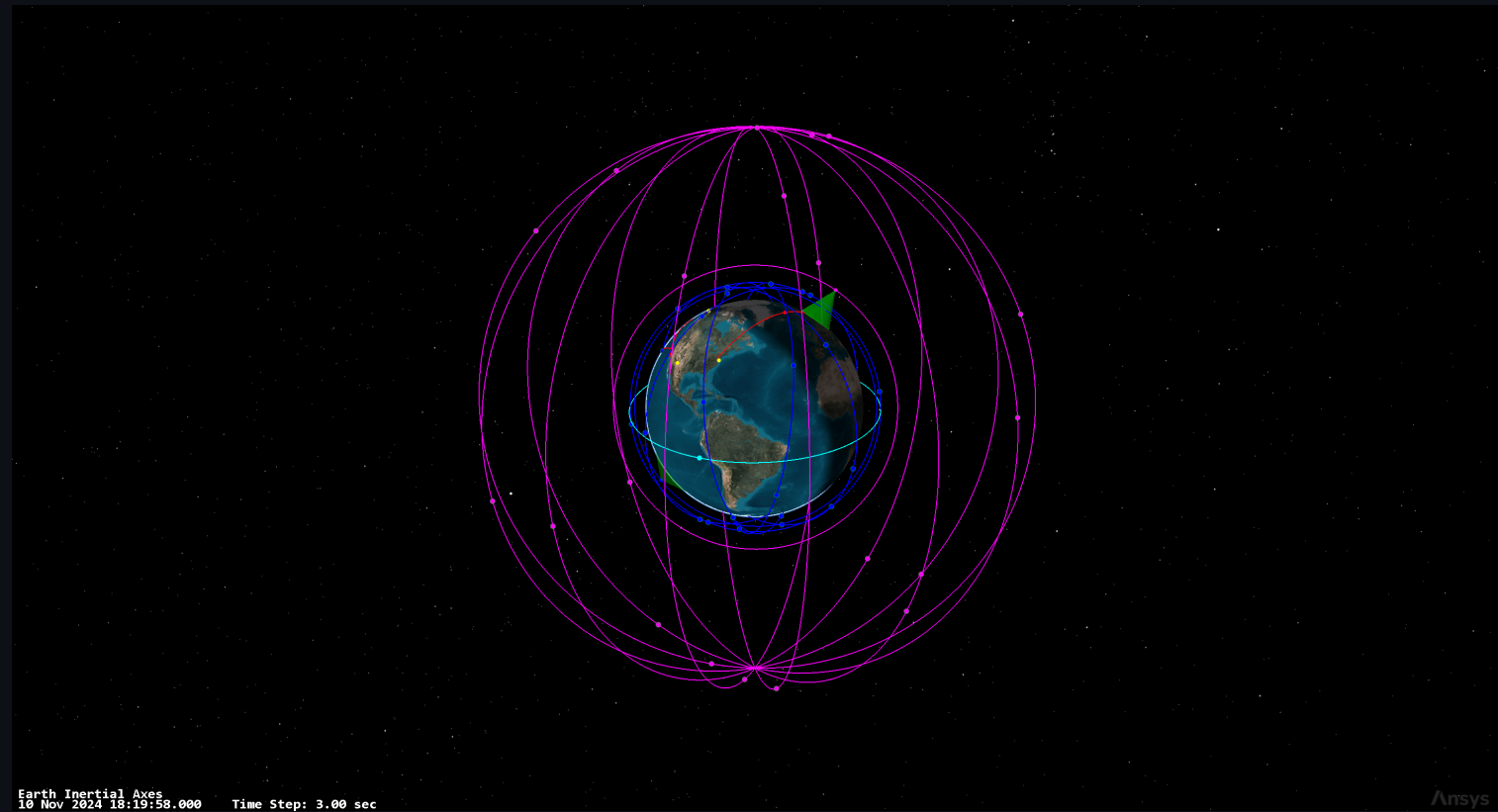
Optimal Average Access Time vs. Number of MEO Satellites



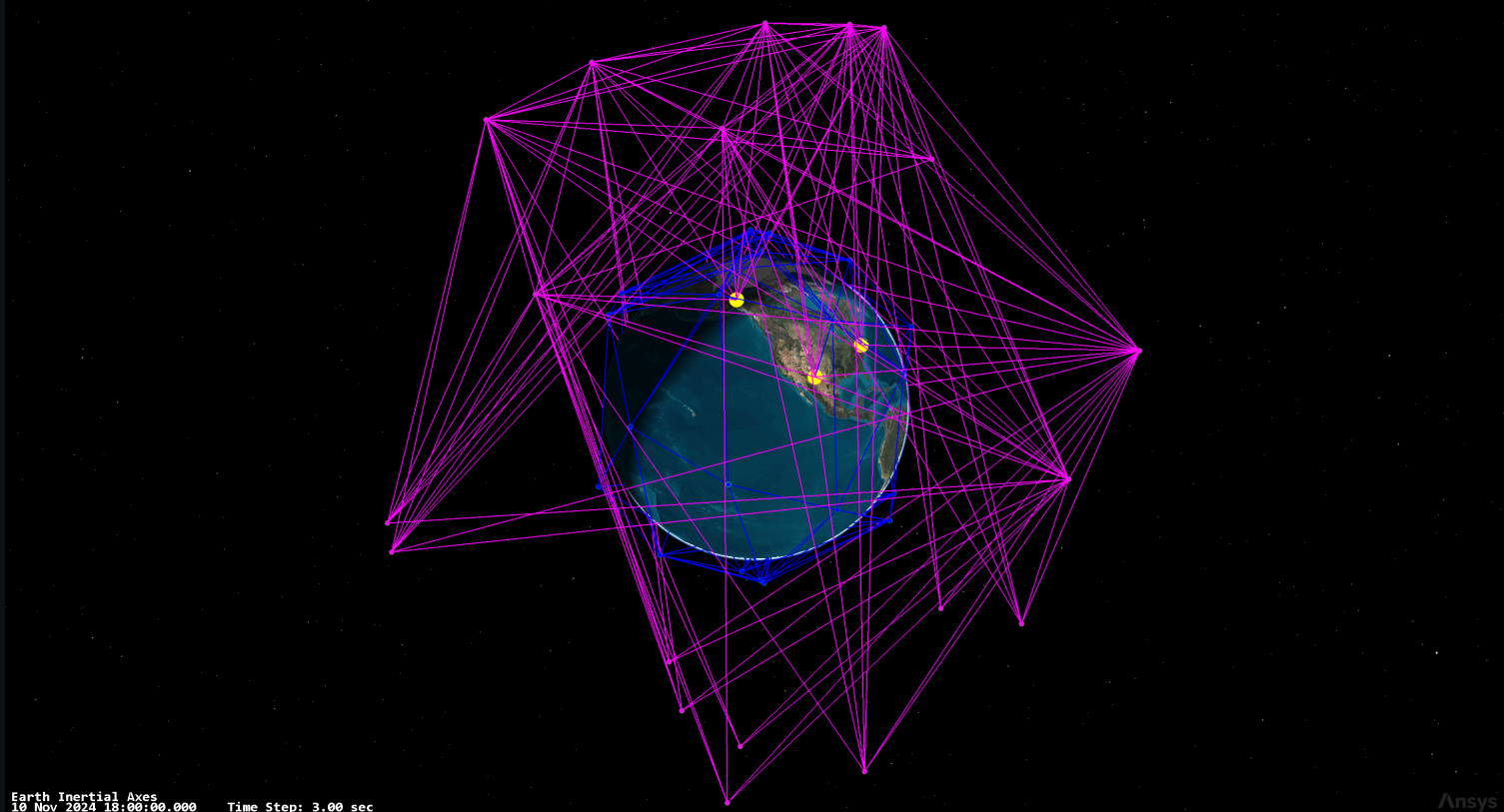
HYPERION



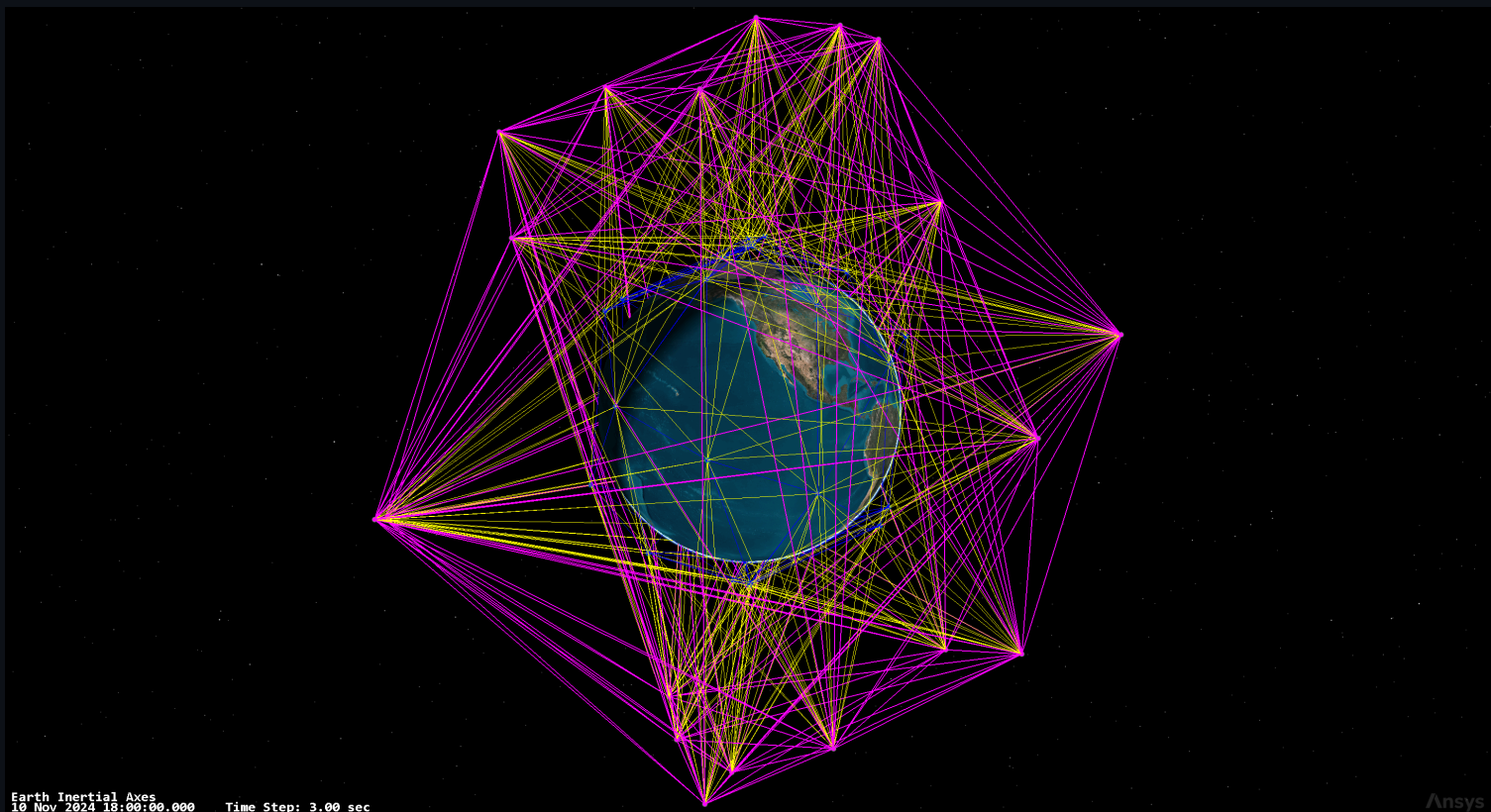
2D representation of HYPERION



3D representation of HYPERION



Instantaneous ground station communication access



Instantaneous satellite communication access

Possible Figures/Animations
Avg revisit vs altitude
Avg revisit vs number of sats
Avg missile response vs number of LEO sats
Avg missile visibility % vs number of MEO sats
Revisit time vs latitude and longitude (heat map)
Full scenario animation
Missile launch with lines of visibility from sensors to missile, sats to sats, and sats to ground stations
Lines of visibility between LEO and MEO, LEO and LEO, MEO and MEO

Traceability to Requirements

System

1. The system shall be capable of detecting hypersonic missile launches within a time frame that allows for effective countermeasure deployment. (Threshold: 30 sec, Objective: 1 sec)
 - Verified by LEO avg missile response
2. The system shall be configured to provide consistent global revisit times supporting timely detection of launches across all latitudes. (Threshold: 10 min, Objective: 1 sec)
 - Verified by revisit time graphs for LEO and MEO
3. LEO satellites shall orbit between altitudes of 500-1000 km.
 - Verified by chosen LEO altitude
4. The LEO constellation shall consist of no more than 32 satellites.
 - Verified by chosen number of LEO sats
5. HYPERION shall be capable of tracking hypersonic missiles throughout their flight path.
 - Verified by MEO avg missile visibility / flight time
6. MEO satellites shall orbit between 2000-10000 km.
 - Verified by chosen MEO altitude
7. HYPERION shall be arranged to provide near-global coverage, ensuring comprehensive observation of all latitudes over a complete orbital period.
 - Verified by revisit time heat map and chosen inclination
8. Satellites shall be capable of in-plane communication
 - Verified by continuous LOS analysis
9. The satellites shall be capable of cross-plane communication.
 - Verified by continuous LOS analysis

Orbits

1. The MEO constellation shall be designed with sufficient satellite density and plane configuration to maintain continuous coverage

- Verified by zero average revisit time for MEO constellation
- 2. The MEO satellites shall be positioned to balance wide-area coverage and low communication latency with the LEO constellation.
 - Verified by zero average revisit time + some average distance to LEO sats?
- 3. The LEO constellation shall have a local time of ascending node (LTAN) staggered to ensure uniform temporal distribution of satellite passes over critical areas.
 - Verified by non-unity phasing factor (f)
- 4. The orbital configuration shall ensure that each LEO satellite maintains continuous line-of-sight (LOS) communication with at least one MEO satellite.
 - Verified by continuous LOS analysis

Next Steps

- More missiles and more accurate missile modeling
- Aircraft missile launches
- Non-zero eccentricity to focus more on high priority regions
- More sensors
- Longer propagation window
- Improve coverage definition granularity